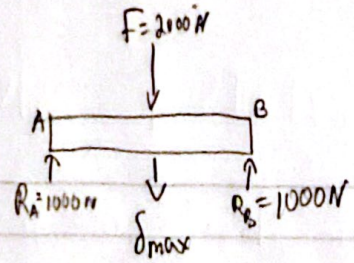
Crosshead Design



Constraints

$\sigma_{max} \leq \frac{\sigma_y}{4}$, $\delta_{max} \leq 1mm$, lead screw distance : 150mm

material: 6061-T6 aluminum, $\sigma_y = 276$ MPa, $E = 68900$ MPa, $width_{CH} = 40$ mm

$F_{max} = 2{,}000N$, Lead screw diameter : 3.658 mm

Stress analysis

$M_{max} = \frac{FL}{4} \rightarrow \frac{2000(150)}{4} =$ **75,000 N·mm**

$\sigma_{allow} = \frac{\sigma_y}{SF} \rightarrow \frac{276}{4} =$ **69 MPa**

$H_{min} \rightarrow \sigma_{max} = \frac{6M}{bh^3} \rightarrow h = \sqrt{\frac{6M}{b\sigma_{allow}}} \rightarrow h = \sqrt{\frac{6(75{,}000)}{(40)(69)}} =$ **12.77 mm** $\approx$ **15 mm**

Deflection analysis

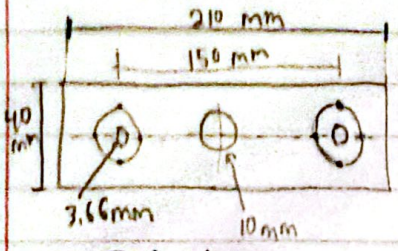
$\delta_{max} = \frac{FL^3}{48EI} \rightarrow \frac{FL^3}{4Ebh^3} \rightarrow h_\delta = \sqrt[3]{\frac{FL^3}{4Eb\delta_{max}}} \rightarrow h_\delta = \sqrt[3]{\frac{(2000)(150)^3}{4(68{,}900)(40)(1)}} =$ **8.44 mm**

$I = \frac{bh^3}{12}$

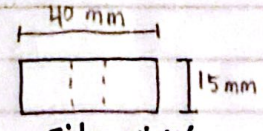
12.77 mm > 8.44 mm ∴ Stress governs

$\sigma_{max} = \frac{6(75{,}000)}{40(15)^2} =$ **50 MPa** $\quad \delta_{max} = \frac{(2000)(150)^3}{4(68{,}900)(40)(15)^3} =$ **0.181 mm** < 1.00 mm

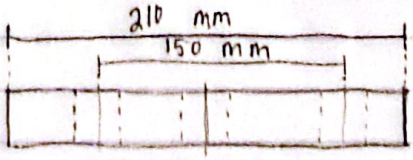
Multiview



Top view



Side view



Front View